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A Omitted Proofs

A.1 Proofs from Section 3

Proof of Lemma 3.2. Recall the definition of alignment utility for skill S_i is the expected log-likelihood under ideal distribution $\mathcal{D}_i$:

$$u_i(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}_i} [\log \pi_\theta(y|x)]$$

The alignment loss is defined as $\Delta u_i(\theta) = u_i(\theta^*) - u_i(\theta)$. Substituting definitions, we obtain

$$\Delta u_i = \mathbb{E}_{(x,y) \sim \mathcal{D}_i} [\log \pi_{\theta^*}(y|x)] - \mathbb{E}_{(x,y) \sim \mathcal{D}_i} [\log \pi_\theta(y|x)]$$

By linearity of expectations we combine terms:

$$\Delta u_i = \mathbb{E}_{x \sim \mathcal{D}_i} \left[\sum_y \mathcal{D}_i(y|x) \log \left(\frac{\pi_{\theta^*}(y|x)}{\pi_\theta(y|x)} \right) \right]$$

Invoking Assumption 1, we have that the aligned policy matches the ideal distribution, $\pi_{\theta^*}(y|x) = \mathcal{D}_i(y|x)$ for all $x \in \text{supp}(\mathcal{D}_i)$. Substituting $\mathcal{D}_i$ with π_{θ^*} , we obtain

$$\Delta u_i = \mathbb{E}_{x \sim \mathcal{D}_i} \left[\sum_y \pi_{\theta^*}(y|x) \log \left(\frac{\pi_{\theta^*}(y|x)}{\pi_\theta(y|x)} \right) \right]$$

where the term inside the expectation is precisely the KL divergence between π_{θ^*} and π_θ . $\square$

Proof of Proposition 3.3. This follows from the second-order Taylor expansion of the KL divergence. The KL divergence, $D_{\text{KL}}(\pi_{\theta^*} || \pi_\theta)$ is a function of θ that is minimized when $\theta = \theta^*$. The gradient of the function at the minimum is zero and the Hessian is the Fisher information matrix. Expanding and taking the expectation over $x \sim \mathcal{D}_i$ gives the desired result. $\square$

A.2 Proofs from Section 4

We first prove the isotropy of the uniform sphere for completeness.

Lemma A.1. *Let $g \sim \text{Unif}(\mathbb{S}^{n-1})$. Then $\mathbb{E}_g[gg^\top] = \frac{1}{n}I_n$*

Proof. Let $M := \mathbb{E}_g[gg^\top]$. For any orthogonal matrix Q , Qg has the same distribution as g by rotational invariance of $\text{Unif}(\mathbb{S}^{n-1})$. Hence,

$$\begin{aligned} M &= \mathbb{E}_g[gg^\top] \\ &= \mathbb{E}_g[(Qg)(Qg)^\top] \\ &= Q\mathbb{E}_g[gg^\top]Q^\top \\ &= QMQ^\top \end{aligned}$$

for any such Q . Thus, M commutes with orthogonal matrices. The only matrices with this property are scalar multiples of the identity, $M = cI_n$ for some $c \in \mathbb{R}$. Taking traces,

$$\text{Tr}(M) = \mathbb{E}_g[\text{Tr}(gg^\top)] = \mathbb{E}_g[\|g\|^2] = 1$$

while $\text{Tr}(cI_n) = cn$. Therefore, $c = 1/n$, proving the claim. $\square$

Moreover, since the Fisher information is a symmetric matrix, we have the following expected form of the quadratic term in our utility function.

Lemma A.2. *Let $g \sim \text{Unif}(\mathbb{S}^{n-1})$ and let $A \in \mathbb{R}^{n \times n}$ be symmetric. Then $\mathbb{E}_g[gAg^\top] = \frac{1}{n}\text{Tr}(A)$.*

Proof. Using $\text{Tr}(uv^\top) = v^\top u$ and the cyclic nature of trace, we have $g^\top Ag = \text{Tr}(g^\top Ag) = \text{Tr}(Agg^\top)$. Taking expectation and applying the linearity of trace and expectation, we obtain $\mathbb{E}_g[g^\top Ag] = \text{Tr}(A\mathbb{E}[gg^\top])$ which by Lemma A.1 finishes the proof. $\square$

We now revisit the equation for change in utility, substituting $\Delta\theta = \eta g$. After taking the expectation and applying Lemma A.2 with $A = F_i(\theta^*)$ we obtain the first bound of the proposition.

We proceed to verify the second claim. First, we prove a standard concentration inequality result on the rank- d projector.

Lemma A.3. *Let $g \sim \text{Unif}(\mathbb{S}^{n-1})$ and let P be any orthogonal projector of rank d . Then for any $\varepsilon \in (0, 1)$,*

$$\Pr \left[\left| \|Pg\|^2 - \frac{d}{n} \right| \geq \varepsilon \cdot \frac{d}{n} \right] \leq 2 \exp(-c\varepsilon^2 d)$$

for constant $c > 0$. In particular, with probability at least $1 - 2e^{-c\varepsilon^2 d}$,

$$(1 - \varepsilon) \frac{d}{n} \leq \|Pg\|^2 \leq (1 + \varepsilon) \frac{d}{n},$$

or $\|Pg\|^2 = \Theta(d/n)$.

Proof. By rotational invariance of g , the distribution of $\|Pg\|^2$ depends only on the rank of P . Thus, we may assume $P = \text{diag}(I_d, 0)$ without loss of generality. Let $z \sim \mathcal{N}(0, I_n)$ and set $g = z/\|z\|$ (ie. a construction of $\text{Unif}(\mathbb{S}^{n-1})$). Define $X := \sum_{j=1}^d z_j^2 \sim \chi_d^2$ and $Y = \sum_{j=d+1}^n z_j^2 \sim \chi_{n-d}^2$, with X and Y independent, and $\|z\|^2 = X + Y$. Then $\|Pg\|^2 = \frac{\|Pz\|^2}{\|z\|^2} = \frac{X}{X+Y}$. We can now invoke standard chi-square Chernoff bounds: for $W \sim \chi_k^2$ and $\varepsilon \in (0, 1)$,

$$\Pr [|W - k| \geq \varepsilon k] \leq 2 \exp(-c_0 \varepsilon^2 k)$$

for a constant $c_0 > 0$. Apply this to X with $k = d$ and to $X + Y \sim \chi_n^2$ with $k = n$ to get that, with probability at least $1 - 2e^{-c_0 \varepsilon^2 d} - 2e^{-c_0 \varepsilon^2 n} \geq 1 - 4e^{-c_0 \varepsilon^2 d}$. This simultaneously yields $(1 - \varepsilon)d \leq X \leq (1 + \varepsilon)d$ and $(1 - \varepsilon)n \leq X + Y \leq (1 + \varepsilon)n$. Combining these results we have

$$\frac{(1 - \varepsilon)d}{(1 + \varepsilon)n} \leq \frac{X}{X + Y} \leq \frac{(1 + \varepsilon)d}{(1 - \varepsilon)n}$$

for $\varepsilon \in (0, 1/2]$ this further becomes

$$(1 - 2\varepsilon) \frac{d}{n} \leq \|Pg\|^2 \leq (1 + \varepsilon) \frac{d}{n}$$

which gives the ultimate result. $\square$

We lastly need to bound the Fisher induced norm by $\|Pg\|^2$.

Lemma A.4. *Let $F \succeq 0$ be symmetric with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ and let P be the orthogonal projector onto the span of the top d eigenvectors of F . Then for any unit vector g ,*

$$g^\top F g \leq \lambda_1 \|Pg\|^2 + \lambda_{d+1} (1 - \|Pg\|^2)$$

In particular, we can further obtain

$$g^\top F g \leq \lambda_1 \|Pg\|^2 + \lambda_1 (1 - \|Pg\|^2) = \lambda_{\max}(F).$$

Proof. Let V_d be the top- d eigenspace of F , and write $g = g_{\parallel} + g_{\perp}$ where $g_{\parallel} := Pg \in V_d$ and $g_{\perp} := (I - P)g \in V_d^{\perp}$. Since V_d and $V_d^{\perp}$ are orthogonal and F is diagonalizable in an orthonormal basis, we have

$$g^\top F g = g_{\parallel}^\top F g_{\parallel} + g_{\perp}^\top F g_{\perp}$$